

closure_evidence_20260923

BOB-197 closure evidence — 2026-09-23

Operator decision (already recorded, 2026-08-26)

“Arm BOBA_API_TOKEN and keep 0.0.0.0, backed by a BOOT-TIME invariant that refuses to start LAN-bound with the token unset.” The token was already generated/armed in .env in a prior session; this item is specifically the still-owed boot-time invariant.

Defect

download-proxy/src/main.py's start_fastapi_server() configured uvicorn with MERGE_SERVICE_HOST defaulting to 0.0.0.0 (LAN-bound) with no check that BOBA_API_TOKEN was armed before serving — the service started open on a default deployment.

Fix

New _refuse_if_lan_bound_and_unarmed(host) function, called after merge_host is resolved and before the uvicorn server starts. Reuses the existing config.proxy._is_loopback helper (no new closed-set check invented). Refuses (via os._exit(1), NOT sys.exit/SystemExit) only when host is non-loopback AND the token is unset/empty. Uses os._exit specifically because start_fastapi_server() runs on a daemon thread — empirically verified SystemExit raised there does NOT terminate the process (only the daemon thread silently dies while

the main thread continues, the original-proxy thread keeps serving on 7186 with no error — exactly the partial-boot anti-pattern §11.4.254 exists to prevent).

Independent verification (coordinator, from clean shell)

```
$ .venv/bin/python -m pytest tests/unit/
test_main.py::TestStartFastapiServerBootTimeLanAuthInvariant
tests/unit/test_main.py::TestRefuseIfLanBoundAndUnarmed tests/
unit/test_main.py::TestMainFastAPIServer -v
14 passed in 2.94s
```

All 4 pre-existing TestMainFastAPIServer tests (which the fix’s real `os._exit(1)` would otherwise have silently killed under an unarmed ambient shell) also pass — confirming the implementing agent’s disclosed regression-fix (arming `BOBA_API_TOKEN` in those tests’ env mocks) is correct.

RED independently reproduced (via git stash, not trusted from report)

```
$ git stash push -m "..." -- download-proxy/src/main.py
$ .venv/bin/python -m pytest tests/unit/
test_main.py::TestStartFastapiServerBootTimeLanAuthInvariant -v
FAILED
test_start_fastapi_server_refuses_lan_bound_default_without_token
  Failed: DID NOT RAISE SystemExit
FAILED
test_start_fastapi_server_refuses_lan_bound_explicit_without_token
2 failed, 2 passed
$ git stash pop # byte-identical restore confirmed via
py_compile
$ .venv/bin/python -m pytest tests/unit/
test_main.py::TestStartFastapiServerBootTimeLanAuthInvariant -q
4 passed
```

Genuine “DID NOT RAISE SystemExit” on the pre-fix code, exactly matching the defect description — not a synthetic/constructed RED.

Golden-FALSE both cases confirmed (part of the same 14-test run)

Loopback+token-unset does not refuse; LAN-bound+token-armed does not refuse (the deployment's actual current happy-path state).

Note found honestly by the implementing agent, not acted on (out of

this item's scope): host was under extreme, unrelated resource contention throughout this dispatch (4 parallel subagents + this coordinator's own gate sweep drove load average to 175+ and swap to full exhaustion) — some invocations appeared to stall from pure CPU/memory starvation, independently confirmed harmless once contention eased. Not a code defect.

Status

Fixed. Closed by coordinator after independent re-verification, including a from-scratch RED reproduction distinct from the implementing agent's own report.